

Gregor Mendel realizes that characteristics (features) of living things are passed on from their parents, in units later called genes.

Friedrich Miescher finds a molecule in the nucleus of pus cells and calls it "nuclein."

Walter Flemming sees structures inside cells, called chromosomes, double before the cell divides, so each new cell gets one copy.

Albrecht Kossel figures out which molecule makes nuclein and names it deoxyribonucleic acid—DNA for short.

1866

1869

1879

1881



Basics of genetics

Using cells from corn plants, Harriet Creighton and Barbara McClintock show that chromosomes carry genetic information.

1931

Rosalind Franklin takes the clearest photograph of DNA yet.

Oswald Avery, Colin MacLeod, and Maclyn McCarty prove that DNA is not just a building block of chromosomes—it also carries genetic information.

A photograph of DNA is produced, showing it has a regular structure.

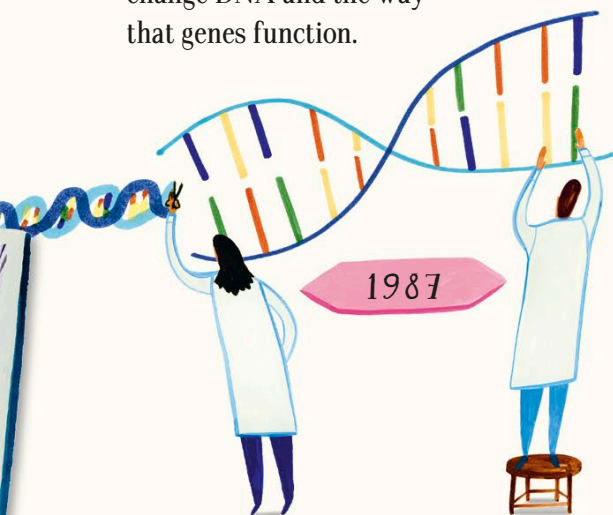
1951

1944

1943

Changing genes

Scientists begin to work on a tool for editing genes. It takes 30 years to develop, but today, scientists can change DNA and the way that genes function.



1987

DNA sequencing

Projects begin to sequence the entire human genome—that's all of our genetic material. It took more than 10 years to read the 3 billion letters in the human DNA code. Today's technology allows anyone's genome to be read accurately.

1990–2003



Genetically modified foods

Tomatoes are the first genetically modified crops to go on sale. Today, gene editing is widely used to give crops useful traits, such as the ability to fend off mold.

1994

Forty years after sickle cell disease was shown to be genetic, gene editing therapy is first used to treat the condition.

DNA from 40,000-year-old Neanderthal bones is studied to find out more about how these prehistoric people are related to modern humans.

Scientists begin sequencing the genomes of microbes that live in and on our bodies to better understand their role in health and disease.

2019

2010

2008

